Sahil Ahamed Shaheena

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.60
Thamarai International School, Thanjavur	Class XII (CBSE)	2023	96.60%
Thamarai International School, Thanjavur	Class X (CBSE)	2021	98.60%

WORK EXPERIENCE

IPHIPI Technologies Pvt. Ltd. -- AI R&D Intern

- Developed and optimized a **production-level wake-word detection system** for always-on TWS earbuds, targeting **<10% FRR @ 0.1 false alarms/hour** under embedded compute and latency constraints.
- Built an **on-the-fly speech/noise data generation pipeline** using diverse multilingual and Indian-accent datasets, with SNR-controlled noise mixing, reverberation, resampling, and partial-wake-word hard negatives.
- Customized **TF-DBPResNet** into a **296.5K-parameter, 11.8M-MAC** KWS model and improved robustness through **curriculum learning, hard-negative generation, and false-alarm replay** using ~100 h of negative-audio inference.
- Diagnosed deployment-condition failures through **indoor/outdoor and SNR-wise FRR analysis**, motivating targeted augmentation; reverberation training reduced the indoor FRR from **11.43% to 8.48%**.
- Performed **INT8 weight QAT and datapath quantization**, validating quantized inference against floating-point benchmarks and tuning the streaming inference strategy under a **<1 s latency constraint**.
- Ported the model to the **Syntiant NDP115**, splitting the network around an unsupported attention-pooling multiply operation and implementing the operation in **C/DSP**; validated Python, simulator, and hardware outputs through bit-matching.

SCHOLASTIC ACHIEVEMENTS

Cleared **National Talent Search Examination** (2020 - 2021) Stage 1

PROJECTS

Arbitrary Precision Arithmetic Calculator in Java (GitHub link)

- Developed a calculator in Java capable of **arbitrary precision arithmetic** operations using **OOP concepts**
- Handles precision with **great accuracy** and bypasses limitations of Java's built-in numeric types
- custom logic for separate **arithmetic integer** and **float classes**

Article summarizer with web-scraper (GitHub link)

- designed and implemented a modular Python application integrating **web scraping** (BeautifulSoup, Requests) and **NLP summarization** (Hugging Face, PyTorch)
- integrated pre-trained **seq2seq model** (BART by default) with **configurable parameters** (e.g. max_length)
- **three-file structure** separating concerns into scraping, summarization, and orchestration for maintainability and scalability

Question-Answering system using RAG: (GitHub link)

- Implemented a QA model using **RAG** and **LangChain** with open-source Hugging Face transformers
- Returns **accurate context-aware answers** using **FAISS** similarity search on uploaded document input
- Utilized NLP methods systematically via **chunking**, **indexing**, **retrieval** and **generation**

Retrieval Augmented Generation-based Large Language models for automated multimodal segmentation

- Deployed BiomedParse (**MLLM** for joint segmentation, detection, and recognition across **9 biomedical imaging modalities**), customizing the original repository to work within Kaggle's compute and platform constraints.
- Engineered a probabilistic cross-view fusion approach (axial, coronal, sagittal) under Prof. Nagarajan's guidance to **mitigate incorrect tumor mask predictions** on non-tumor slices, improving segmentation reliability.
- **Reduced false positive rates** in 3D tumor segmentation to **zero** on 100k non-tumor slices data by integrating and

reconciling probability maps across multiple views, leveraging cross-plane inconsistencies to eliminate spurious voxel predictions.

EXTRACURRICULAR

Meltwater Hangman Hackathon (Github link)

- Developed a character-level **CANINE-S Transformer** from scratch in PyTorch to solve 6-strike Hangman, modeling masked word states and 26-letter action probabilities over **225K+ training words**.
- Engineered a two-stage training pipeline combining **biased Monte-Carlo state generation and batched self-play fine-tuning**, with frequency-weighted soft-target cross-entropy and reproducible checkpoint-based training.
- Built an offline Hangman simulator and evaluation pipeline for **250K test words**, with iterative state updates, strike-aware inference, validation tracking, and automated Kaggle submission generation.
- Achieved approximately **66% win rate on a held-out test set**, with modular data generation, training, evaluation, and inference components optimized for **2× NVIDIA T4 GPUs**.

KLA Hackathon (Github link)

- Developed a **supervised grayscale image restoration pipeline** to reconstruct 256×256 clean images from 128×128 low-resolution images corrupted by additive Gaussian noise and speckle noise.
- Modeled the unknown image degradation process as a **composition of downsampling and noise transformations** with varying strengths and ordering, using paired ground-truth/degraded samples to guide preprocessing and training.
- **Built** and **trained a deep image-to-image restoration model** for joint denoising and 2× super-resolution, optimizing pixel-level reconstruction against the original ground-truth images.
- Implemented **evaluation and data-processing pipelines** for .npy float32 image data, using PSNR, SSIM, and reconstruction error to quantify restoration quality.

Volunteered at Cleanliness and Donation Drives at NSS

Participated in multiple elocution competitions at district level

SKILLS

Programming Languages: C, C++, Java, HTML, CSS, Python

Platforms & Tools: VS Code, Jupyter, Linux, Solid Edge, Kaggle, GitHub, Google Colab, Vim-Bash

Python Libraries: Numpy, Matplotlib, Scikit, Pytorch, Pandas, Streamlit, Keras

RELEVANT COURSES

Computer Science: Introduction to Programming in C, Data Structures & Algorithms, Computer Architecture, Database Management Systems, Discrete Math, Operating Systems I, Theory of Computation, Software Development Fundamentals

Mathematics: Calculus I & II, Linear Algebra, Probability and Random Variables, Group Theory, Statistics, Transform Techniques, Differential Equations, Real Analysis, Vector Calculus

Artificial Intelligence & Machine Learning: Foundations of Machine Learning, Concentration Inequalities, Reinforcement Learning, Deep Learning

POSITIONS OF RESPONSIBILITY

Internship Coordinator, Office of Career Services

Placement Coordinator, Office of Career Services

Content & Curation Coordinator, TEDxIITHyderabad

Multimedia Coordinator, MILAN

Production Core, Behind The Lens